

Computer Networks Final Exam Questions and Answers

Azarbaijan Shahid Madani University - Second Semester of Academic Year 2025-2026

Question 1

A sender node has 5 frames available. Send these 5 frames to the destination using the one-way ARQ method. Consider the window size to be 4. Assume that in the first transmission, frames number 2 and 4 do not reach the destination and are dropped along the way. Fully explain and demonstrate the usage of timeouts and ACK frames. (15 points)

Answer:

Given the sender window size of 4, we use the **Go-Back-N (GBN)** protocol. In this method, if a frame is lost, the receiver discards subsequent frames, and the sender retransmits all unacknowledged frames after the time-out period expires.

Question 2

Send the data string 1001101 to a destination using the CRC method. Assume that during transmission, the second bit from the left has encountered an error. Using the aforementioned method, detect and reveal the error. Use the string 1001 for error calculations at both source and destination. (20 points)

Answer:

A) Calculations at the Sender Side: The divisor is 1001, which consists of 4 bits. Therefore, we must append $4 - 1 = 3$ zeros to the end of the original data:

$$\text{Data} = 1001101000$$

Now we perform binary division (XOR):

Step	Sender (Source)	Receiver (Destination)	Description and Status
1	Send Frame 1	-	Window: [1, 2, 3, 4]
2	Send Frame 2	-	Frame 2 is lost in transit.
3	Send Frame 3	-	
4	Send Frame 4	-	Frame 4 is lost in transit.
5	-	Receive Frame 1 Send ACK 2	Frame 1 received successfully, requesting Frame 2.
6	Receive ACK 2	-	Window shift: [2, 3, 4, 5]
7	Send Frame 5	-	Sending new frame.
8	-	Receive Frame 3	Receiver expects Frame 2. Frame 3 is discarded and ACK 2 is re-sent.
9	-	Receive Frame 5	Frame 5 is discarded and ACK 2 is resent.
10	Frame 2 Timeout	-	Frame 2 timer expires.
11	Retransmit 2, 3, 4, 5	-	According to GBN, all unacknowledged frames are resent.
12	-	Receive Frames 2 to 5	Successful reception and final ACK sent.

1001101000 $\div$ 1001

1001

00001010

1001

00110

0000

01100

1001

0101 (Remainder or CRC)

Since we need a 3-bit remainder, the value 101 is appended to the data as the CRC.

Sent Data = 1001101101

B) Calculations at the Receiver Side (Assuming Error Occurs): Problem

assumption: The second bit from the left has an error (changes from 0 to 1):

Received Data = 1101101101

Division at the receiver side:

$$\begin{array}{r} 1101101101 \div 1001 \\ \underline{1001} \\ 01001 \\ \underline{1001} \\ 00001011 \\ \underline{1001} \\ 001001 \\ \underline{1001} \\ 0000 \end{array} \quad (\text{Remainder is non-zero in final stages})$$

Since the remainder at the end of the division is not zero, the receiver detects an error in the received frame and discards it.

Question 3

In the RED buffer management method, the buffer length is 14, the first 5 packets are accepted 100 percent, and the last 3 packets are definitely dropped (the buffer is FIFO). Calculate the RED random packet drop formulas and describe the algorithm's performance for 14 consecutive packets with time intervals t . Packets placed in the queue leave the buffer with a time interval of $5t$. (15 points)

Answer:

Determining RED Algorithm Parameters: Since the first 5 packets are accepted 100%, the minimum queue threshold is:

$$\text{min_threshold} = 5$$

Also, because the buffer length is 14 and the last 3 packets are definitely dropped, the maximum queue threshold is obtained as:

$$\text{max_threshold} = 14 - 3 = 11$$

Random Drop Probability Formula (P_b):

$$P_b = \frac{\max_p \times (avg - \min_threshold)}{\max_threshold - \min_threshold} = \frac{\max_p \times (avg - 5)}{11 - 5} = \frac{\max_p \times (avg - 5)}{6}$$

Packet Arrival and Departure Scheduling Table: Assuming one arrival every t seconds and departures occurring every $5t$:

Time	Operation	Queue Count	Acceptance Status according to RED
$1t$	Arrival of Packet 1	1	Guaranteed acceptance ($avg \leq 5$)
$2t$	Arrival of Packet 2	2	Guaranteed acceptance
$3t$	Arrival of Packet 3	3	Guaranteed acceptance
$4t$	Arrival of Packet 4	4	Guaranteed acceptance
$5t$	Arrival of Packet 5 Departure of Packet 1	4	Guaranteed acceptance. Departure occurs after arrival 5.
$6t$	Arrival of Packet 6	5	Lower threshold boundary. Start of random drop probability (P_b).
$7t$	Arrival of Packet 7	6	Random drop probability (P_b) increases.
$8t$	Arrival of Packet 8	7	Checked with random probability.
$9t$	Arrival of Packet 9	8	Checked with random probability.
$10t$	Arrival of Packet 10 Departure of Packet 2	8	Departure occurred, count remained constant.
$11t$	Arrival of Packet 11	9	Checked with random probability.
$12t$	Arrival of Packet 12	10	Checked with random probability.
$13t$	Arrival of Packet 13	11	Reaching the upper threshold boundary.
$14t$	Arrival of Packet 14	12	Above maximum threshold (11). Guaranteed drop.

Question 4

A PC and a server are located in a subnet as shown below. Describe the TCP Handshaking steps between these two computers. Specify and describe the role of the IP addresses and MAC addresses written in this figure during this Handshaking. (20 points)

Answer:

Due to being in the same subnet, these two devices do not need a router to communicate and directly establish physical communication via MAC addresses at the data link layer.

Role of Addresses:

- **IP Address (Layer 3):** Used for logical identification of source and destination at the network level and remains constant throughout the process.
- **MAC Address (Layer 2):** Used for physical delivery of frames directly between the network interface cards of the two devices in the local network.

TCP 3-Way Handshake Steps:

1. **First Step (SYN):** The PC sends a synchronization request packet to the DNS server.
 - IP Addresses: Source 130.12.12.5 / Destination 130.12.0.15
 - MAC Addresses: Source 4eef4ff800d2 / Destination 77ff4c08800c
2. **Second Step (SYN-ACK):** The DNS server receives the packet, acknowledges it (ACK), and sends its own synchronization request (SYN) to the PC.
 - IP Addresses: Source 130.12.0.15 / Destination 130.12.12.5
 - MAC Addresses: Source 77ff4c08800c / Destination 4eef4ff800d2
3. **Third Step (ACK):** The PC acknowledges the server's message and sends the final ACK packet to the server. With this, a stable TCP connection is established.
 - IP Addresses: Source 130.12.12.5 / Destination 130.12.0.15
 - MAC Addresses: Source 4eef4ff800d2 / Destination 77ff4c08800c